

Computational Microelectronics

Lecture 8

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L8

Issues

- We have written a code to calculate I_D .
 - ~~1) Electron continuity equation is analytically solved.~~
 - 2) MOS equation and continuity equation are separately treated.
 - 3) Mobility is assumed to be constant.
 - 4) Short-channel MOSFET
 - 5) Source/drain regions
 - 6) Transient effect
 - 7) Multiple devices (Can we calculate the VTC of a CMOS inverter?)

Convergence problem

- Considering the real space is good.
 - However, it does not solve the convergence problem.
 - Update of ϕ_n was too large.
- Eventually, we must couple the MOS equation and the continuity equation.
 - It is today's topic.

Solution vector

- We have two equations.

- MOS equation:

$$V_G - V_{fb} = -\frac{Q_s(\phi_s, \phi_n)}{C_{ox}} + \phi_s$$

- Electron continuity equation:

$$\frac{d}{dx} \left(Q_n(\phi_s, \phi_n) \mu_n \frac{d\phi_n}{dx} \right) = 0$$

- Two variables are ϕ_s and ϕ_n : $\begin{bmatrix} \phi_n(x_0) \\ \vdots \\ \phi_n(x_{N_x-1}) \end{bmatrix} \rightarrow \begin{bmatrix} \phi_s(x_0) \\ \phi_n(x_0) \\ \vdots \\ \phi_s(x_{N_x-1}) \\ \phi_n(x_{N_x-1}) \end{bmatrix}$

Nonlinear + multiple variables

- We solve a nonlinear equation with the Newton method.
 - Now, we have several (discretized) equations, $2 \times N_x$.
 - Extension to multivariate cases
- Consider a set of coupled nonlinear equations.
 - Three variables, ϕ_1 , ϕ_2 , and ϕ_3 .
 - Three equations read
$$\begin{aligned} F_1(\phi_1, \phi_2, \phi_3) &= \phi_2 - 2\phi_1 - e^{\phi_1} = 0 \\ F_2(\phi_1, \phi_2, \phi_3) &= \phi_3 - 2\phi_2 + \phi_1 - e^{\phi_2} = 0 \\ F_3(\phi_1, \phi_2, \phi_3) &= -2\phi_3 + \phi_2 - e^{\phi_3} + 4 = 0 \end{aligned}$$
 - How can we find the solution by using a numerical method?

Derivation (1)

- Of course, we did it. But, repeat it again.

- To find a set of ϕ_1 , ϕ_2 , and ϕ_3 , which satisfies

$$\begin{bmatrix} F_1(\phi_1, \phi_2, \phi_3) \\ F_2(\phi_1, \phi_2, \phi_3) \\ F_3(\phi_1, \phi_2, \phi_3) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- For the moment, we don't know the solution.

- We can make a guess, $\phi^0 = [\phi_1^0 \quad \phi_2^0 \quad \phi_3^0]^T$.

$$r = \begin{bmatrix} F_1(\phi_1^0, \phi_2^0, \phi_3^0) \\ F_2(\phi_1^0, \phi_2^0, \phi_3^0) \\ F_3(\phi_1^0, \phi_2^0, \phi_3^0) \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- By correcting our guess, we want to make a residual vector r small.

Derivation (2)

- Update vector, $\delta\phi = [\delta\phi_1 \quad \delta\phi_2 \quad \delta\phi_3]^T$

– We hope that $\phi^0 + \delta\phi$ is the real solution.

$$\begin{bmatrix} F_1(\phi_1^0 + \delta\phi_1, \phi_2^0 + \delta\phi_2, \phi_3^0 + \delta\phi_3) \\ F_2(\phi_1^0 + \delta\phi_1, \phi_2^0 + \delta\phi_2, \phi_3^0 + \delta\phi_3) \\ F_3(\phi_1^0 + \delta\phi_1, \phi_2^0 + \delta\phi_2, \phi_3^0 + \delta\phi_3) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

– By using the linearization, we can have the following equation:

$$\begin{bmatrix} F_1(\phi_1^0, \phi_2^0, \phi_3^0) \\ F_2(\phi_1^0, \phi_2^0, \phi_3^0) \\ F_3(\phi_1^0, \phi_2^0, \phi_3^0) \end{bmatrix} + \begin{bmatrix} \partial F_1 / \partial \phi_1 & \partial F_1 / \partial \phi_2 & \partial F_1 / \partial \phi_3 \\ \partial F_2 / \partial \phi_1 & \partial F_2 / \partial \phi_2 & \partial F_2 / \partial \phi_3 \\ \partial F_3 / \partial \phi_1 & \partial F_3 / \partial \phi_2 & \partial F_3 / \partial \phi_3 \end{bmatrix} \begin{bmatrix} \delta\phi_1 \\ \delta\phi_2 \\ \delta\phi_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Derivation (3)

- Introducing the Jacobian matrix J and the residual vector r
 - In terms of these quantities, the update vector, $\delta\phi$, is written as:

$$J\delta\phi = -r$$

- After $\delta\phi$ is calculated, the improved solution vector, ϕ^1 , is constructed:

$$\phi^1 = \phi^0 + \delta\phi$$

- In every Newton step, r and $\delta\phi$ are monitored. When these vectors approach the null vector, we have a better solution.

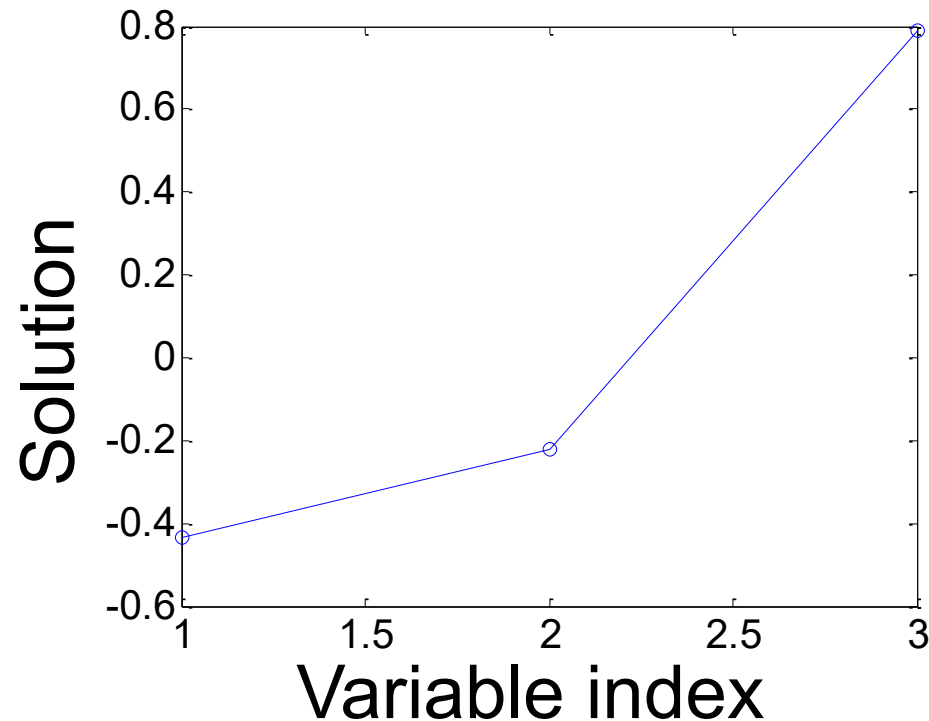
Jacobian matrix

- Calculation of the Jacobian matrix is the key step.
 - For the k -th step with a solution vector of ϕ^k , the Jacobian matrix of our example reads

$$J = \begin{bmatrix} -2 - \exp \phi_1^k & 1 & 0 \\ 1 & -2 - \exp \phi_2^k & 1 \\ 0 & 1 & -2 - \exp \phi_3^k \end{bmatrix}$$

Solution vector

- The solution is $[-0.4352 \quad -0.2233 \quad 0.7884]^T$.
 - Even such a small problem cannot be solved analytically. (It's a reason why we study this course.)



MOSFET under a fixed ϕ_n profile (1)

- Instead of MOS equation + electron continuity,
 - MOS equation + external ϕ_n profile (Covered in L5)
 - But, now, we have $2 \times N_x$ variables.
 - The MOS equation is local in the x coordinate.

$$F_{\phi_s} = V_G - V_{fb} + \frac{Q_s(\phi_s, \phi_n)}{C_{ox}} - \phi_s = 0$$

- Therefore, the following entries of the Jacobian matrix are also local.

$$\frac{\partial}{\partial \phi_s} F_{\phi_s} = \frac{1}{C_{ox}} \frac{\partial}{\partial \phi_s} Q_s(\phi_s, \phi_n) - 1, \quad \frac{\partial}{\partial \phi_n} F_{\phi_s} = \frac{1}{C_{ox}} \frac{\partial}{\partial \phi_n} Q_s(\phi_s, \phi_n)$$

- Just like $\frac{\partial}{\partial \phi_s} Q_s(\phi_s, \phi_n)$, $\frac{\partial}{\partial \phi_n} Q_s(\phi_s, \phi_n)$ can be calculated.

MOSFET under a fixed ϕ_n profile (2)

- External ϕ_n profile

$$F_{\phi_n} = \phi_n - \phi_{n,external} = 0$$

- The following entries of the Jacobian matrix are local.

$$\frac{\partial}{\partial \phi_s} F_{\phi_n} = 0, \quad \frac{\partial}{\partial \phi_n} F_{\phi_n} = 1$$

- Then, a part of the set of equations looks like:

$$\begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ \dots & \frac{1}{C_{ox}} \frac{\partial}{\partial \phi_s} Q_s(\phi_s, \phi_n) - 1 & \frac{1}{C_{ox}} \frac{\partial}{\partial \phi_n} Q_s(\phi_s, \phi_n) & \dots \\ \dots & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \times \begin{bmatrix} \vdots \\ \delta \phi_s \\ \delta \phi_n \\ \vdots \end{bmatrix} = - \begin{bmatrix} \vdots \\ F_{\phi_s} \\ F_{\phi_n} \\ \vdots \end{bmatrix}$$

Thank you!